



Sizing Design a Hybrid PV-Wind System with Battery Storage using IHOGA Simulator

Seddik Baala ^{a,*}, Brahim Berbaoui ^b, Abdelkader Harrouz ^c, Brahim Dahhou ^d

^{a,b,c} Department of Electrical Engineering, Laboratoire du Développement Durable et Informatique (LDDI), University of Adrar, Algeria

^d Department of Electrical Engineering, University of Mohamed Boudiaf Oran, Algeria

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ABSTRACT

Renewable energy sources offer a viable solution to meet energy demands while simultaneously reducing environmental pollution from conventional power plants. However, the inherent variability of weather conditions renders reliance on a single energy source inadequate. Therefore, a hybrid renewable energy system (HRES), incorporating both solar and wind energy, along with battery storage, is necessary. This study investigates the techno-economic aspects of a hybrid system in an isolated area in Adrar, Algeria, employing the IHOGA (Improved Hybrid Optimization by Genetic Algorithms) software to yield noteworthy results. The system design includes a 4.4 kW photovoltaic array, a 0.92 kW wind turbine, a 56.16 kWh battery bank, and a 3000 W inverter. The loads are proposed for the region, which has an average annual energy consumption of 2554.9778 kWh/year. The simulation results reveal the total system cost Net Present Cost (NPC) over a 25-years lifespan, level cost of energy plus CO₂ emissions, and environmental impacts.

1. INTRODUCTION

Increasing energy demands and the pressing need to mitigate environmental pollution have led to a growing interest in renewable energy sources. Hybrid renewable energy systems (HRES), which integrate multiple renewable sources such as solar, wind, and biomass, offer a sustainable solution to meet the energy needs of both urban and isolated areas (Thirunavukkarasu et al, 2020). As traditional energy systems, reliant on fossil fuels, face challenges such as resource depletion and pollution, the transition to HRES becomes increasingly crucial (Reciouui et al., 2022). This paper focuses on the optimal design of a hybrid renewable energy system for an isolated area (24.932, 0.605) in Adrar, as

* Corresponding author, E-mail address: seddik.baala@univ-adrar.edu.dz

Tel : + 213 696809643

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shown in Figure1, utilizing IHOGA software for precise modelling and simulation. The reliance on a singular renewable source often proves inadequate due to variability in weather conditions; therefore, the combination of multiple renewable sources enhances system reliability and efficiency (Ganguly et al., 2017). The development of HRES is particularly vital for remote locations where grid access is limited, providing these areas with sustainable and self-sufficient energy (Fares et al., 2022). In addition to addressing energy shortages, HRES can significantly reduce carbon emissions, contributing to a cleaner environment (Niveditha & Rajan Singaravel, 2022). Advanced modelling tools like IHOGA enable the effective sizing and configuration of system components to meet specific energy demands while optimizing costs (Hoarcă et al., 2023; Hussain et al., 2024; Lujano-Rojas et al., 2023). This study explores the design process and evaluates the economic and environmental benefits associated with hybrid systems, drawing on recent advancements in renewable energy technologies. The methodology employed herein aligns with findings from previous research, such as the work by (Hoarcă et al., 2023), which highlights the potential of using simulation tools like HOMER and IHOGA for designing hybrid power systems effectively. By analysing case studies and existing literature, this paper aims to contribute to the knowledge surrounding hybrid renewable energy systems and their application in isolated regions.

2. METHODOLOGY

This research utilizes a techno-economic design and optimization framework to create an independent hybrid energy system incorporating photovoltaic, wind, and battery technologies, specifically directed toward fulfilling the electricity needs of an isolated area in the Adrar province of Algeria, chosen for this case study. The main tool used in the application of this framework was the sophisticated simulation software IHOGA (Improved Hybrid Optimization of Genetic Algorithms). The data relevant to renewable energy resources, solar irradiation, and wind speed for the area chosen were gleaned from NASA databases, in addition to devising an estimated electrical load profile representing consumption patterns throughout a whole year. In addition, the input data included the technical and economic parameters of the major elements: photovoltaic (PV) solar panels, wind turbines, storage batteries, and inverters, specifying their capital outlays, operating life, and operation and maintenance (O&M) expenses (Shamachurn, 2021). The main optimization goal was determined as minimizing the system's Net Present Cost (NPC), subject to a strict reliability constraint that required keeping the Loss of Power Supply Probability (LPSP) very close to zero. The program used a Genetic Algorithm (GA) to perform an iterative search for the best system configuration, considering both the number and size of its components, and the battery's State of Charge (SOC) control strategy. The outcomes obtained from the optimization process, deserving of careful analysis, include the optimal system configuration, the annual energy contribution of each source, and major economic parameters like the Levelized Cost of Energy (LCOE) and the payback period.

3. IHOGA SOFTWARE ALGORITHM

This study utilized simulation software called IHOGA, which is widely applied across both academic and industrial contexts for hybrid energy system optimization and design. IHOGA is a powerful tool with the ability to simulate standalone (off-grid) and grid-connected systems that utilize multiple energy sources such as photovoltaic arrays, wind turbines, and battery energy storage systems (BESS) consisting of both lead-acid and lithium-ion technologies together with auxiliary (Hoarcă et al., 2023; Recioui et al., 2022). The effectiveness of this software lies in using a Genetic Algorithm (GA) to carry out multi-objective optimization. The main goal is to determine the optimum system configuration that strives to minimize economic parameters, including the Net Present Cost (NPC), CO₂ emissions and

the Levelized Cost of Energy (LCOE), while at the same time aiming to achieve maximum levels of reliability, commonly measured based on the Loss of Power Supply Probability (Ganguly et al., 2017). IHOGA simulations evaluated configurations across energy demand scenarios to balance reliability, cost-effectiveness, and sustainability.

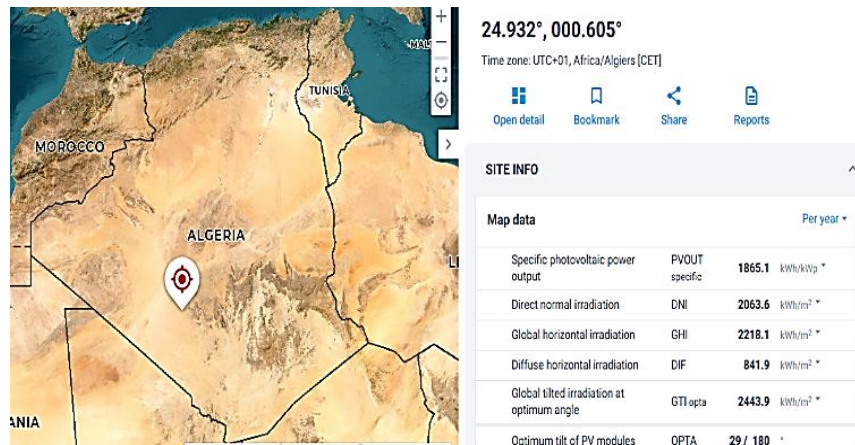


Fig 1. Location of the area chosen

4. COMPONENT OF THE HYBRID RENEWABLE ENERGY SYSTEM

The system examined in this study is a hybrid configuration comprising photovoltaic generators, wind turbines, and battery storage, intended to supply AC load needs in an isolated area of Adrar. To address load demand, wind turbines and photovoltaic (PV) panels convert available wind and solar energy resources at the designated site into electricity (Chen et al., 2024). The battery bank functions as an energy storage system, providing power during periods of insufficient electricity generation and storing surplus energy when production exceeds (Bonkile & Ramadesigan, 2022; Fathima & Palanisamy, 2015). The system configuration is depicted in Figure 2. The components data is shown in Table 1.

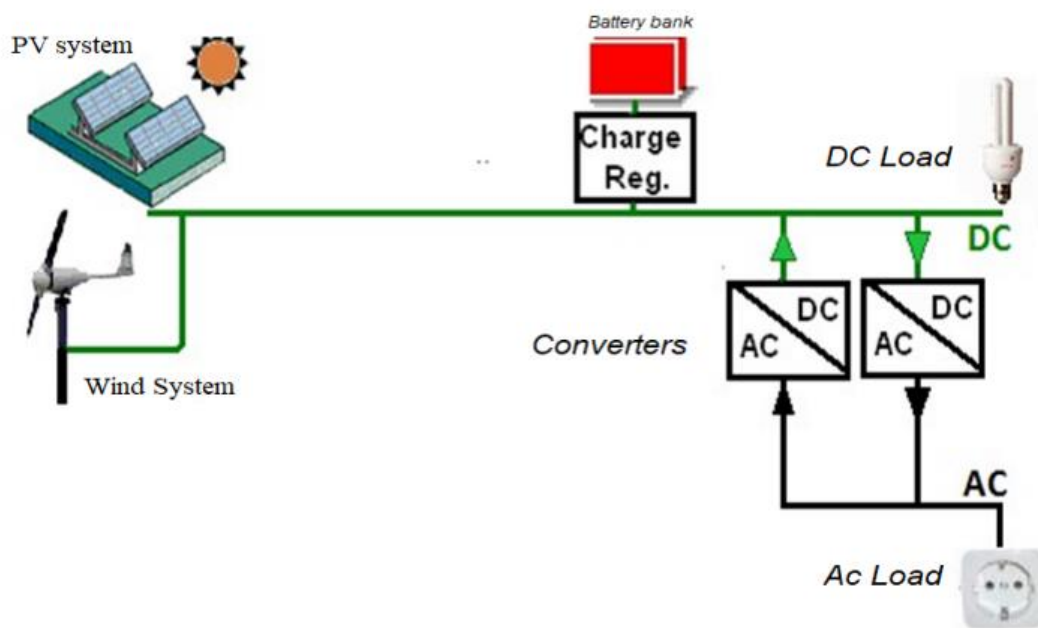


Fig 2. Modal of the hybrid renewable energy system in IHOGA

Table 1. System Components Specification

Components	Specification	Total power/ Energy
PVmod.aSi12 -Schott: ASI100	3series*8parallel *100 W	2.4 kW
Battery. OPZS Hawker: TVS-5	24series*1parallel*390 Ah	18.72 kWh
Wind Turban Southwest: Whisper100	1*925 W (for wind 14m/s)	0.92 kW
Inverter Generic: 3000	1*3000 Watt	3 KW

4.1 Photovoltaic Generator (GPV) System

The photovoltaic panel converts solar radiation into electricity, while IHOGA software can upload solar radiation data via NASA's site based on the geographical location (Adrar, Algeria).

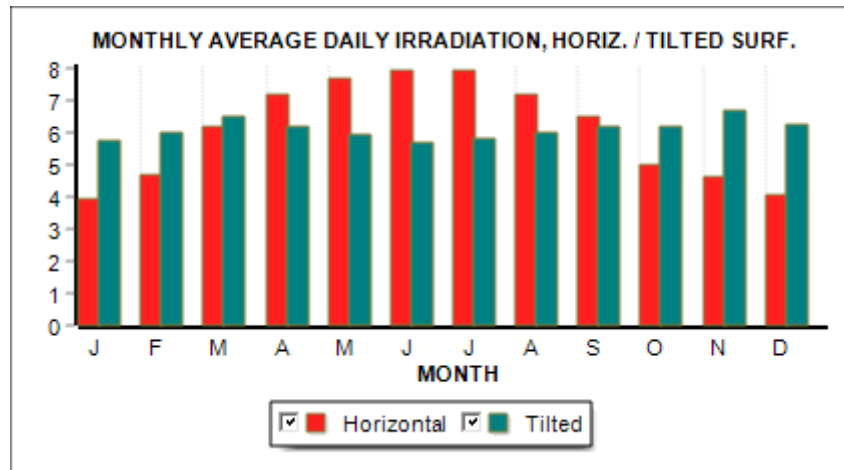


Fig 3. Solar radiation for one month

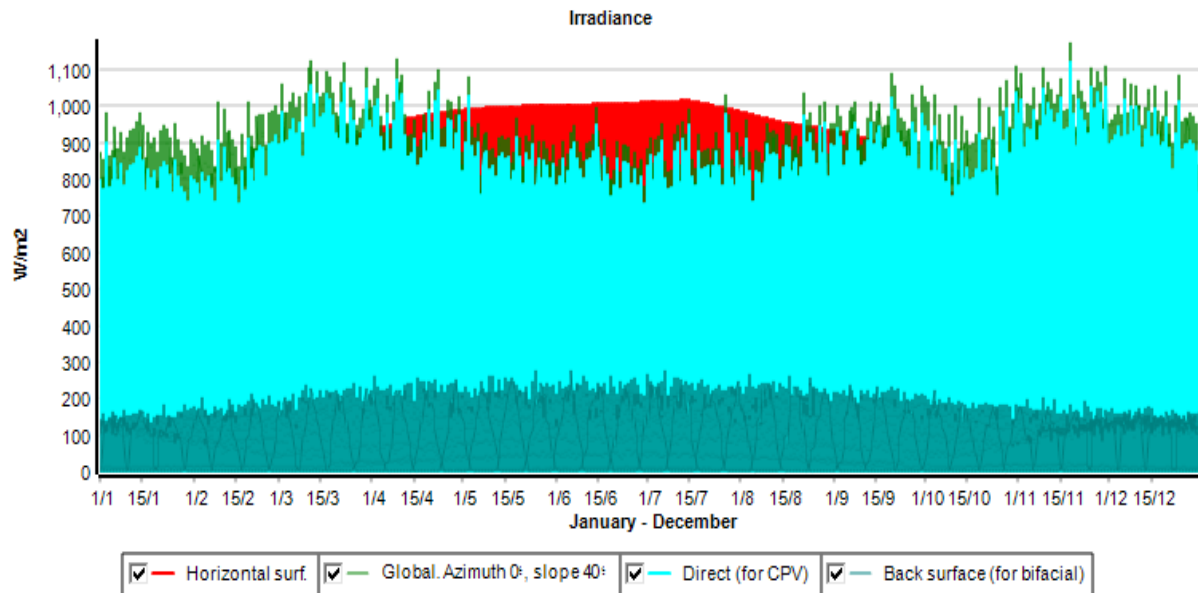


Fig 4. Total Solar radiation for one year.

Figure 3 illustrates the solar radiation profile for both horizontal and tilted surfaces for one year period in isolated area (24.932, 0.605) of Adrar. In this study, we utilize aSi12-Schott: ASI100 type photovoltaic panels, each with a nominal voltage of 12V and a power output of 100W. Figure 4 illustrates the total solar radiation in the same area for one year. The acquisition cost of the PV panels is

54 \$ per unit. The lifetime of the PV panels is 25 years (Araoye et al., 2024; Nogueira et al., 2014). The operations and maintenance (O&M) cost is assumed to be 1.43 \$ per year (Bacha et al., 2024).

4.2 Wind energy system

The average wind speed for this area was obtained from NASA's website using IHOGA software. Figure 5 illustrates the wind speed at this location over one year. The wind turbine used in this project is the Southwest Whisper 100. Figure 6 shows power changes concerning wind speed for this type.

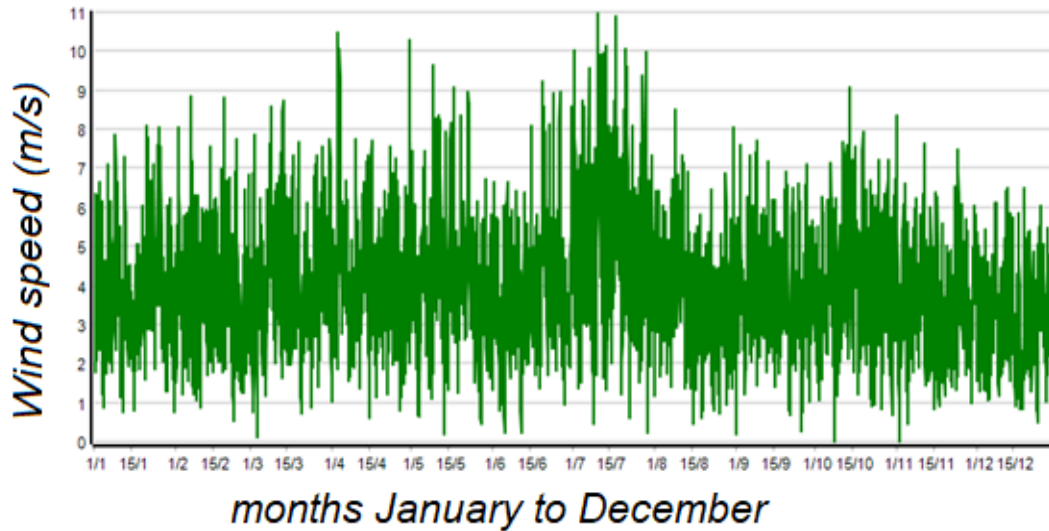


Fig 5. Wind speed of Adrar during one year

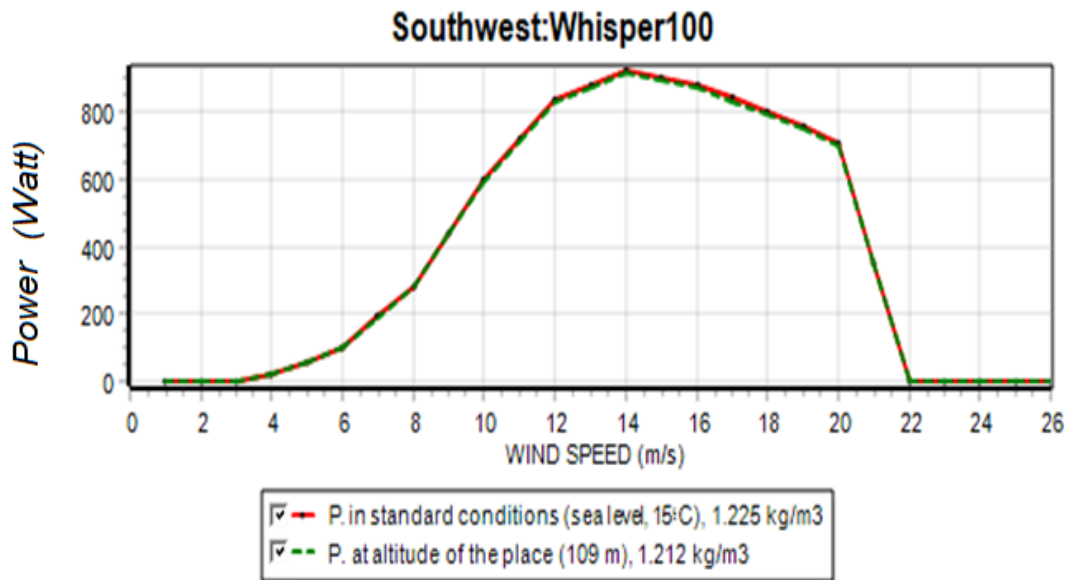


Fig 6. Characteristics of the Southwest Whisper 100 Wind Turbine

4.3 Energy Storage System (Battery Bank)

In this study, we used OPZS-Hawker: TVS-5 battery type of 390 Ah. The battery's acquisition cost is 25.72 \$ per unit, and its annual operating and maintenance costs are 0.26 \$ per unit. The battery lasts for eighteen years. Each unit has a nominal voltage of 2V, and according to the manufacturer's

specifications, the battery bank's minimum state of charge (SOC) is 20%. Figure 7 provides the battery bank's detailed properties.

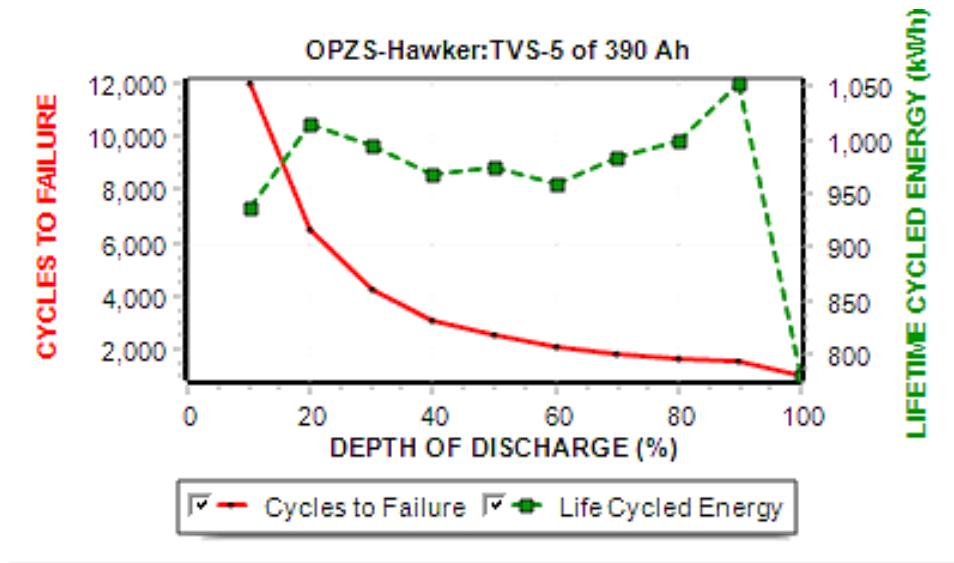


Fig 7. OPZS-Hawker: TVS-5 battery characteristics

4.4 DC-AC Converter (Inverter)

The Generic: 3000 Inverter type is relied upon in this study which has 3000VA power and a 10-year lifespan.

4.5 Load Profile

Figure 8 presents the load profile for January in one day for the proposed small town in an isolated area of Adrar

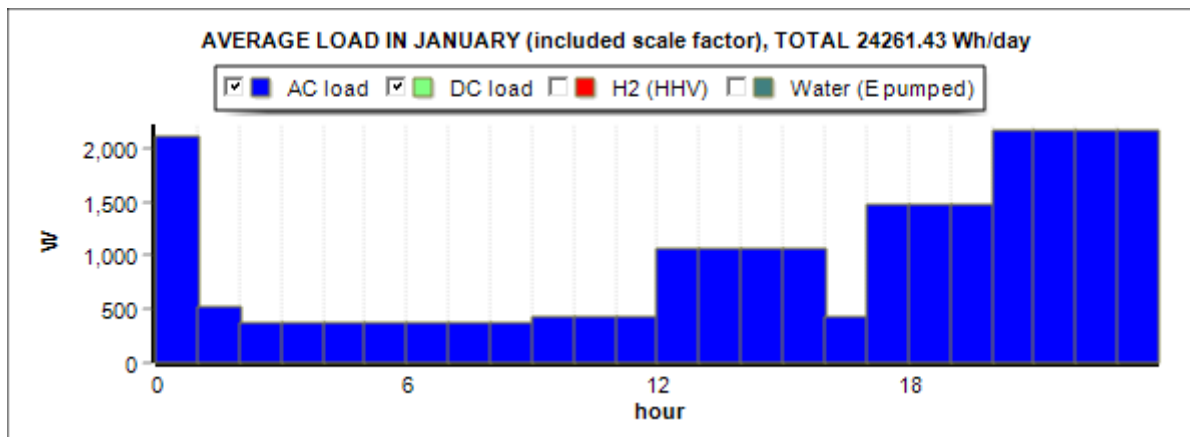


Fig 8. load profile in one day

5. RESULTS AND DISCUSSION

The results obtained from the simulations for the different Sensitivity analyses are displayed in Figure 9.

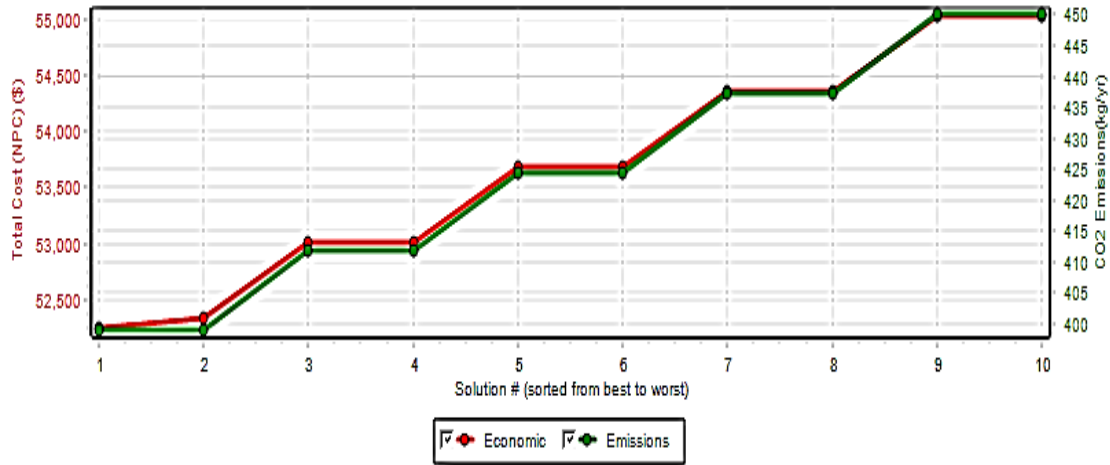


Fig 9. The sensitivity analysis results

5.1 Costs

- Initial Investment: 8,907.27 \$.
- Total System Costs (NPC), 25-years lifetime: 10,650.00 \$.
- Level Cost of Energy: 0.12 \$/kWh.

5.2 Distribution of NPC

Figure 10 illustrates the system's cost analysis. It highlights the following results: The battery had the highest cost (NPC): 4,579.50 \$ (representing 43% of the system cost); the inverter and the auxiliary components cost (NPC) was 1810.5 \$ (17%); the PV generator cost (NPC) was 2,662.5 \$ (25%); and the wind turbine cost (NPC) was 1,597.5 \$ (15%).

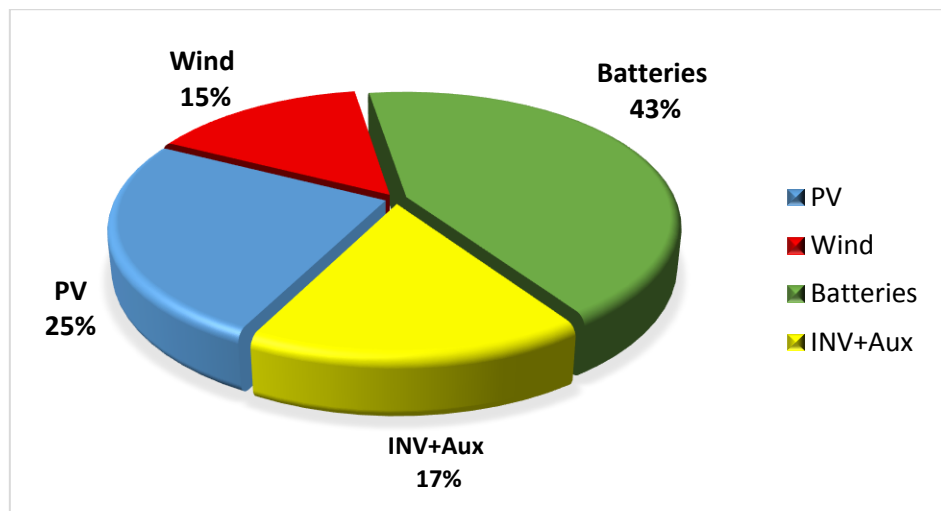


Fig 10. Cost summary of the selected system

5.3 Energy Balance for 1 year (kWh/year)

A year's worth of data on the system's energy balance is presented in Figure 11. The system simulation gave the following results:

- Overall Load Energy: 2555 kWh/yr.
- Export Energy: 185 kWh/yr
- Energy delivered by PV generator: 3100 kWh/yr
- Energy delivered by Wind Turbines 450 kWh/yr
- Energy charged by Batteries: 1750 kWh/yr.
- Energy discharged by Batteries: 1480 kWh/yr.

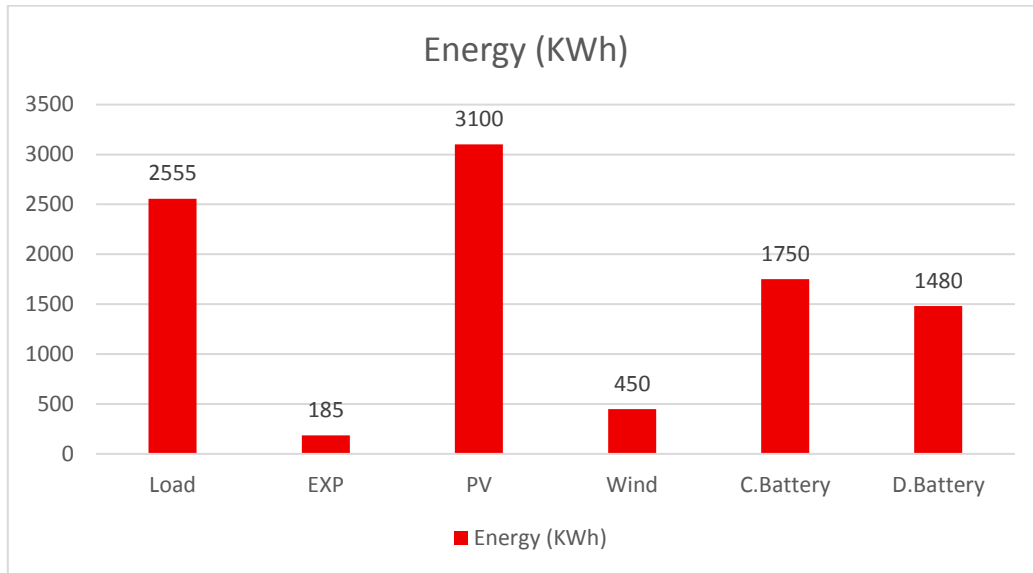


Fig 11. Energy balance of the selected system for one year

5.4 Other Results

- Batteries' Lifetime: 12.5 years
- Total CO₂ emissions: 215kgCO₂/yr

The sensitivity analysis findings presented in Figure 9 disclosed that batteries are the most significant factor in determining the economic viability of the hybrid system, contributing 43% to the NPC. Their lifespan of 12.5 years is less than the expected nominal lifespan of 18 years because charge-discharge cycles and depth of discharge occur frequently, hence making the NPC highly responsive to variations in battery specs. On the other hand, changes in wind energy contribution have a comparatively low impact because its annual contribution, about 450 kWh, is not as significant as that from photovoltaic arrays, about 3100 kWh.

Cost allocation in Figure 10 emphasizes the huge economic role of batteries, followed by photovoltaic systems (25%), inverters (17%), and wind power (15%). This is necessitated by the need for maintenance policies that aim to extend the lifespan of these components. The yearly energy assessment in Figure 11 shows an excess production of about 3550 kWh/year compared to the demand of 2555 kWh/year, with a round-trip efficiency of storage of about 84.6%. This excess ensures load coverage during unfavourable conditions; however, it also underscores storage inefficiencies. In conclusion, the results validate that improving the efficiency of the storage system and the optimization of net energy management are vital parameters in the accomplishment of a lower LCOE and the sustainability of the system at the specified site.

6. CONCLUSION

This study successfully demonstrates the techno-economic feasibility of a standalone hybrid power system (consisting of photovoltaic, wind, and battery subsystems) for electricity generation in a remote area of Adrar in Algeria, as established through an exhaustive optimization process with the IHOGA software. The optimization resulted in a systematic reduction in component sizes, especially in the energy storage system, thus reducing the Levelized Cost of Energy (LCOE) to 0.12 \$/kWh. While the cost is higher than urban tariff levels, it is still very competitive and economically sound compared to the extension of the expensive national grid to remote areas. Besides economic considerations, the system has a reliable and renewable source of energy with an environmental benefit of reducing greenhouse gas emissions. Future studies might involve new technologies of storing and advanced control methods for improving system efficiency and increasing the service period of the equipment.

REFERENCES

- Araoye, T. O., Ashigwuike, E. C., Mbunwe, M. J., Bakinson, O. I., & Ozue, T. I. (2024). Techno-economic modeling and optimal sizing of autonomous hybrid microgrid renewable energy system for rural electrification sustainability using HOMER and grasshopper optimization algorithm. *Renewable Energy*, 229, 120712. <https://doi.org/10.1016/j.renene.2024.120712>
- Bacha, B., Ghodbane, H., Dahmani, H., Betka, A., Toumi, A., & Chouder, A. (2024). Optimal sizing of a hybrid microgrid system using solar, wind, diesel, and battery energy storage to alleviate energy poverty in a rural area of Biskra, Algeria. *Journal of Energy Storage*, 84, 110651. <https://doi.org/10.1016/j.est.2024.110651>
- Bonkile, M. P., & Ramadesigan, V. (2022). Effects of sizing on battery life and generation cost in PV–wind battery hybrid systems. *Journal of Cleaner Production*, 340, 130341. <https://doi.org/10.1016/j.jclepro.2021.130341>
- Chen, Q., Kuang, Z., Liu, X., & Zhang, T. (2024). Optimal sizing and techno-economic analysis of the hybrid PV-battery-cooling storage system for commercial buildings in China. *Applied Energy*, 355, 122231. <https://doi.org/10.1016/j.apenergy.2023.122231>
- Fares, D., Fathi, M., & Mekhilef, S. (2022). Performance evaluation of metaheuristic techniques for optimal sizing of a stand-alone hybrid PV/wind/battery system. *Applied Energy*, 305, 117823. <https://doi.org/10.1016/j.apenergy.2021.117823>
- Fathima, H., & Palanisamy, K. (2015). Optimized Sizing, Selection, and Economic Analysis of Battery Energy Storage for Grid-Connected Wind-PV Hybrid System. *Modelling and Simulation in Engineering*, 2015, 1–16. <https://doi.org/10.1155/2015/713530>
- Ganguly, P., Kalam, A., & Zayegh, A. (2017). Design an optimum standalone hybrid renewable energy system for a small town at Portland, Victoria using iHOGA. 2017 Australasian Universities Power Engineering Conference (AUPEC), 1–6. <https://doi.org/10.1109/AUPEC.2017.8282487>
- Hoarcă, I. C., Bizon, N., Șorlei, I. S., & Thounthong, P. (2023). Sizing Design for a Hybrid Renewable Power System Using HOMER and iHOGA Simulators. *Energies*, 16(4), 1926. <https://doi.org/10.3390/en16041926>
- Hussain, S., Sharma, S. K., & Lal, S. (2024). Feasible synergy between hybrid solar PV and wind system for energy supply of a green building in Kota (India): A case study using iHOGA. *Energy Conversion and Management*, 315, 118783. <https://doi.org/10.1016/j.enconman.2024.118783>

- Lujano-Rojas, J., Dufo-López, R., & Domínguez-Navarro, J. A. (2023). Optimization of renewable energy systems by genetic algorithms. In *Genetic Optimization Techniques for Sizing and Management of Modern Power Systems* (pp. 247–281). Elsevier. <https://doi.org/10.1016/B978-0-12-823889-9.00007-2>
- Niveditha, N., & Rajan Singaravel, M. M. (2022). Optimal sizing of hybrid PV–Wind–Battery storage system for Net Zero Energy Buildings to reduce grid burden. *Applied Energy*, 324, 119713. <https://doi.org/10.1016/j.apenergy.2022.119713>
- Nogueira, C. E. C., Vidotto, M. L., Niedzialkoski, R. K., De Souza, S. N. M., Chaves, L. I., Edwiges, T., Santos, D. B. D., & Werncke, I. (2014). Sizing and simulation of a photovoltaic-wind energy system using batteries, applied for a small rural property located in the south of Brazil. *Renewable and Sustainable Energy Reviews*, 29, 151–157. <https://doi.org/10.1016/j.rser.2013.08.071>
- Reciou, A., Benaissa, N., & Dekhandji, F. Z. (2022). Hybrid Renewable Energy System Optimization using iHOGA. *Algerian Journal of Signals and Systems*, 7(3), 99–108. <https://doi.org/10.51485/ajss.v7i3.167>
- Shamachurn, H. (2021). Optimization of an off-grid domestic Hybrid Energy System in suburban Paris using iHOGA software. *Renewable Energy Focus*, 37, 36–49. <https://doi.org/10.1016/j.ref.2021.02.004>
- Thirunavukkarasu, M., & Sawle, Y. (2020). Design, analysis and optimal sizing of standalone PV/diesel/battery hybrid energy system using HOMER. *IOP Conference Series: Materials Science and Engineering*, 937(1), 012034. <https://doi.org/10.1088/1757-899X/937/1/012034>